BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **Data Analysis Report**

Project ID: R26-DS-006

Project Title: AI – BASED ADAPTIVE VR MEDITATION SYSTEM

1. Introduction

1.1. Background:

Psychological stress is an important variable that impacts mental wellness, productivity, emotional control, and relaxation effectiveness. Conventional approaches for detecting stress rely heavily on self-reporting questionnaires and retrospective feedback after a session, which cannot adequately support adaptive systems. Advances in wearable sensing technology, speech-based emotion analytics, and virtual reality applications make it possible to evaluate stress through physiological, acoustic, and behavioral data continuously.

Among various physiological biomarkers, Heart Rate Variability (HRV) is the most frequently applied feature in studies assessing psychological stress since it captures the activity of the autonomous nervous system and sympathetic-parasympathetic balance [1]. In this study, RR intervals are obtained from the ECG signal in the WESAD dataset to construct our HRV feature set [2]. In the wearable device, RR intervals can be derived similarly as Inter-Beat Interval (IBI) from the PPG sensor such as MAX30102. Nonetheless, individual differences exist in physiological responses, and they also vary with time according to biological rhythms. For this reason, Cosinor Rhythmometry is applied in our physiological stress inference component to estimate an expected physiological rhythm based on MESOR, amplitude, and acrophase, followed by calculating deviation from this rhythm [3].

The Virtual Reality (VR) experience offers an immersive medium to promote relaxation as it enables users to immerse themselves into restorative environments, such as those depicting nature scenery, tranquil visual environments, and breathing environments. From existing research, VR-based relaxation and HRV biofeedback have been found to be effective in reducing stress and promoting psychological wellbeing [4], [5]. Nonetheless, most existing meditation applications in VR have been static and generic in design. This poses a problem since relaxation is a personal experience, and people differ in terms of sensory preference, their response to stress, susceptibility to cybersickness, among others [6]–[8].

Sound is another essential component of VR meditation, as it affects breathing rate, arousal, sense of calmness, and attention. The current meditative techniques usually incorporate generic sounds or non-adaptive tracks despite differences in users' preference of ambience, types of instruments, brightness, tempo, ability to accept novelty, and stimulatory needs. Studies regarding relaxing music and mindfulness sound reveal that musical organization and stimulus selection impact relaxation quality [9], [10]. Thus, the present work introduces an adaptive audio system based on confidence information to correlate individual user

preference, estimated stress level, and physiological confidence with safe and real-time manipulation of both meditative and environmental sound tracks. Reinforcement learning methods like PPO, TD3, and SAC algorithms may be used to adaptively control parameters since they can learn policies through interaction data [11]–[13].

In addition to physiological and audio adaptive methods, voice could serve as a biomarker for measuring stress. Cognitive stress would cause changes in features such as the pitch variance, speech speed, vocal tension, breathing patterns, and energy levels of the voice [14]–[16]. In contrast to the wearable biosensors, voice would be recorded using a conventional microphone, allowing its utilization as a practical modality in detecting stress before and after each session. Voice stress detectors have typically adopted binary classification models and emotional speech data generated by actors in the past. For overcoming this limitation, emotion2vec—a speech-based emotion embedding learning algorithm—which maps emotion embeddings from voice into continuous-valued stress scores within the framework of Valence-Arousal has been utilized within the project [17], [18]. The publicly available RAVDESS and CREMA-D databases, consisting of emotional speech data, are employed for evaluating the voice stress model and ablation studies [19], [20]. The results are then cross-validated against HRV-based stress estimates, while a separate anomaly detection algorithm is utilized for recognizing abnormal meditation sessions [21].

In essence, the suggested AI-Based Adaptive VR Meditation System integrates HRV-based physiological stress assessment, Cosinor rhythm-based modeling, voice-based cognitive stress assessment, audio adaptation, and reinforcement-learning based optimization of VR parameters to enable the development of a personalized feedback-looped meditation system capable of stress detection, validation of user state through multiple channels, and adaptation of the VR environment for relaxation purposes.

1.2. Research Problem:

There are various limitations associated with the current approaches to stress inference and VR-based meditation. First, many models of stress detection from HRV data directly infer stress based on raw HRV values while ignoring natural interindividual differences and time dependencies due to physiological rhythms. Thus, changes in HRV typical for natural rhythms will be incorrectly recognized as HRV changes related to stress response. Rhythm-sensitive stress inference should compare observed HRV dynamics with the predicted physiological rhythm pattern using techniques such as cosinor rhythmometry [1], [3].

Another limitation associated with adaptive VR environments is related to fixed decision making rules and/or adaptations. Adaptive control of an environment is based on specific rules which do not take into account individual differences of physiological rhythms and personal sensory preferences and factors affecting relaxation and risk of cybersickness. Reinforcement learning can help solve these problems but still has inherent problems of cold

start, delay in adaptation, proper reward design, etc. [6]–[8]. In particular, the wrong decision regarding physiological state may be made leading to incorrect adaptation.

Thirdly, adaptation to audio in virtual reality meditation is still lacking. Although audio properties like tempo, loudness, ratio of ambient sounds and music, brightness, spaciousness, and novelty affect relaxation, many applications rely on either background music or static soundscapes. The current approaches usually lack physiological confidence in their control schemes, meaning that there is the risk of overfitting to noisy or less confident stress estimation. Therefore, an audio control algorithm suitable for real-time deployment and accounting for physiological confidence is needed [9]–[13].

Fourthly, most voice-based stress assessment algorithms only determine whether a speech sample is stressed or not. The resulting binary value is not sufficient to assess the difference between stress levels at the beginning and the end of a session. Furthermore, current voice recognition systems have been trained on acted emotional speech samples like RAVDESS and CREMA-D, which do not necessarily capture spontaneous speech [19], [20]. Voice-based stress detection also lacks physiological validation, so voice scores must be validated against HRV-based stress values.

Lastly, many of the current VR meditation systems lack an automated assessment of session effectiveness in terms of their ability to lower stress levels. If the system does not effectively reduce stress or if stress response is abnormal, then this is not detected automatically and, hence, an anomaly detection system should be created to detect such abnormal and ineffective sessions [21].

Consequently, the overall research question being tackled by this study is:

How could AI-powered adaptive VR meditation incorporate rhythm-aware inference of physiological stress, confidence-aware audio adaptation, cognitive stress measurement using vocal input, cross-modal validation, and reinforcement-learning-powered VR parameter tuning to provide a personalized and reliable user experience?

1.3. Objectives:

It Number	Objective	Objective number
IT22191038	For extracting and pre-processing the RR/IBI interval data obtained from either an ECG or PPG-based physiological signal for HRV-based stress analysis.	1

	For employing Cosinor rhythmometry to calculate rhythm parameters, including MESOR, amplitude, and acrophase, for the establishment of personalized physiological baseline.	2
	For developing HRV, circadian, and Cosinor residual features based on the differences between real physiological measures and the rhythm baseline.	3
	For training and testing of machine learning and deep learning methods for four-level stress classification (relaxed, mild stress, moderate stress, and high stress).	4
	For evaluating the stress inference engine's robustness, generalization performance, and potential for real-time application based on benchmark data sets and prototype-level physiological data.	5
IT22231314	Collect user preference inputs and stress classification outputs required for adaptive VR environment initialization.	1
	Design a rule-based initializer to generate safe starting VR parameters based on stress level and user preferences.	2
	Implement an RL-based adaptation mechanism to adjust VR environment parameters during relaxation sessions.	3
	Evaluate the adaptive component using simulated and pilot- session interaction logs, reward trends, stress-reduction indicators, and user comfort feedback.	4
IT22172396	To design a long-term audio preference representation and continuous target mapping for personalized VR meditation audio control.	1
	To construct a simulated-user environment that models hidden preference targets, tolerance widths, stress proxy, and confidence-aware state transitions.	2
	To define and compare controller conditions for gap-focused evaluation, including generic static, personalized static, confidence-aware rule-based, and PPO-based adaptive control	3

	To prepare Unity-compatible outputs such as safe action targets, session logs, and dashboard-ready traces for runtime validation of the selected controller.	4
IT22621788	To develop a pre-session ambient environment assessment mechanism that analyzes surrounding acoustic noise levels and determines whether the recording environment is suitable for voice-based stress assessment.	1
	To implement a voice-based cognitive stress scoring pipeline using the emotion2vec foundation model, producing continuous stress scores on a 0–10 scale using the Valence-Arousal psychological framework.	2
	To develop a pre/post session stress comparison mechanism and cross-modal validation framework that compares voice-derived stress scores against HRV-based physiological stress estimates.	3
	To design and train an autoencoder-based anomaly detection model to automatically identify unusual or ineffective meditation sessions by measuring deviation from normal stress reduction patterns.	4

2. Data Exploration

2.1. Data Collection

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

The primary dataset employed in this component is the WESAD dataset, containing signals from wearable sensors based on various affective states [2]. Chest ECG signal was selected in this component as it offers high accuracy in detecting R-peak in order to extract RR intervals. ECG signal was initially filtered using NeuroKit2 for R-peak detection, computing RR intervals, artifact removal, and labeling [22].

As a second source of data for cross-dataset experimentations and transfer learning purposes, the SWELL HRV dataset is utilized. SWELL offers HRV features collected from individuals experiencing work-induced stresses, such as no stress, interruptions, and time pressure conditions [23].

Regarding the direction of the prototype design, MAX30102 PPG is taken into account as an example of wearable signal sources. It offers raw IR PPG signals along with timestamps that could be translated into pulse peaks, IBI values, HRV features, and stress prediction results.

Along with the WESAD dataset, the SWELL-KW dataset was also utilized in this study in order to analyze and evaluate the stress prediction model. In the SWELL-KW dataset, there is an abundance of physiological and behavioral data gathered while the subjects were engaged in office tasks with varying working conditions such as neutral, time-pressure, and interruptions. This dataset consists of physiological signals associated with stress that can be used in stress detection using machine learning techniques.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

The component does not depend on a pre-existing public dataset. Instead, data is collected and generated through interaction between the user, the mobile application, the physiological stress classification component, the Unity VR environment, and the RL agent. The main data streams are user preference data, stress classification inputs, VR environment state logs, RL state-action-reward transition logs, synthetic user profiles, and post-session feedback.

During early implementation, synthetic user profiles are used to simulate different relaxation preferences and stress-response patterns. This enables testing of the RL loop before real user testing. In later evaluation stages, pilot sessions with university students will be used to collect real interaction data. Each session will generate repeated records of the current observation, selected action, updated VR state, reward value, and session feedback.

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

The component currently uses four data categories. The first is a structured long-term preference schema defined for meditation audio personalization, covering sound style, instrument set, tempo, mood, intensity, balance, brightness, novelty tolerance, dissonance tolerance, rhythm preference, nature sound level, reverb, and volume. The second is a synthetic user cohort generated from that schema for prototype offline analysis. Each synthetic user is mapped to a hidden continuous target profile and tolerance widths so that adaptive control can be tested before real user collection. The third category is simulated transition data generated during environment rollouts, containing state, action, safe filtered action, reward, and controller-mode traces. The fourth category is prototype runtime logging produced in Unity, where stress and confidence inputs, audio parameter outputs, and safety states can be visualized and exported for later analysis.

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

The voice-based stress assessment module uses three categories of data. The primary training data consists of three publicly available acted speech emotion datasets: RAVDESS, CREMA-

D, and TESS; which provide labeled emotional speech recordings for model evaluation and ablation study experiments. The second category is pre-trained model knowledge from emotion2vec_base_finetuned, which was pre-trained on over 40,000 hours of diverse emotional speech data across multiple languages and recording conditions. The third category is session-level stress data generated during system operation, including pre-session and post-session stress scores, cross-modal validation outputs, and session feature vectors used for autoencoder anomaly detection training.

For the ablation study comparing wav2vec2, emotion2vec, and the finalized classification approach, the three acted speech datasets were combined into a single corpus of 9,227 samples, split into training (80%), validation (10%), and test (10%) sets using speaker-independent stratification to prevent data leakage.

2.2. Dataset Description:

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Data source	Description	Resource	Size	Key attributes
WESAD Dataset	Multimodal physiological dataset for stress and affect detection	Public Dataset (Kaggle)	15 subjects	ECG signals, RR intervals, HRV features, timestamps, stress labels
SWELL-KW Dataset	Workplace stress dataset collected under different office working conditions	Public Dataset (Kaggle)	25 participants	Mean RR, SDNN, RMSSD, pNN50, LF, HF, LF/HF, SD1, SD2, stress labels
MAX30102 PPG stream	Prototype-level wearable sensor input for future real-time stress inference.	ESP32-S3 + MAX30102 sensor	Real-time stream	timestamp_ms, raw IR PPG value

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

Data Source	Description	Resource	Size	Key Attributes
User preference data	Preferences collected before a VR session through the mobile app.	Kotlin mobile app and cloud/backend storage	Small initially; grows with number of users	Preferred brightness, color tone, scene density, visual motion tolerance, fog preference, session comfort settings
Stress classification data	Stress class received from the physiological stress detection component.	Teammate's HR/HRV-based ML component	One class per interval or session checkpoint	Low, moderate, high, very high stress labels; timestamp; user/session ID
VR environment state logs	Current values of controllable VR visual parameters.	Unity VR forest environment	Generated continuously during sessions	Brightness, fog density, sky tone, object density, animation/motion level, breathing guide visibility
RL interaction logs	State-action-reward transition records produced by the RL loop.	Unity RL module / local or backend log storage	Large over repeated episodes	Observation vector, action vector, reward, next observation, episode number, timestep
Synthetic user profiles	Artificial profiles are used to test the agent before real pilot sessions.	Manually designed simulator rules	3-10 profiles initially; expandable	Preference cluster, stress sensitivity, expected relaxation curve, tolerance limits
Session feedback data	User feedback collected after VR exposure.	Mobile app questionnaire	One record per session	Relaxation rating, comfort rating, nausea/discomfort score, preferred changes, overall satisfaction

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

Data source	Description	Resource	Size	Key attributes
Long-term audio preference schema	Structured preference definition used to personalize meditation audio and encode user-specific targets.	Component design specification / questionnaire design	13 fields; 66,134,880 theoretical profile combinations	Sound style, instrument subset, tempo, mood, intensity, balance, brightness, novelty, dissonance, rhythm, nature, reverb, volume
Synthetic user cohort	Generated users sampled from the schema and mapped into continuous control targets and tolerance widths.	Prototype simulator	1,500 users (1,000 train / 250 validation / 250 test)	Discrete preferences, continuous targets, tolerance widths, user-type diversity
Simulated RL transition records	Environment rollouts used to score candidate controllers and export training-ready transitions.	Generated by simulated environment	Variable-length per episode	State vector, action deltas, safe action, reward, confidence, controller mode
Unity prototype runtime logs	Session-level traces produced by the playback prototype for runtime inspection and dashboard validation.	Generated by Unity prototype	Variable-length per session	Timestamp, stress proxy, confidence, final safe action, audio outputs, mute/fallback state

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Data Source	Description	Resource	Size	Key Attributes
RAVDESS	Acted emotional speech from 24 professional North American English actors.	Zenodo (CC-BY)	1,440 speech files; 24 actors; 8 emotions	Emotion label (calm, happy, sad, angry, fearful, surprise, disgust, neutral), intensity, actor ID, gender
CREMA-D	Crowd-sourced emotional multimodal actors dataset with diverse ethnic backgrounds.	Publicly available	7,442 clips; 91 actors; 6 emotions	Emotion label (angry, disgust, fear, happy, neutral, sad), intensity, actor ID, age, ethnicity
TESS	Toronto Emotional Speech Set recorded by two female speakers.	Publicly available	2,800 clips; 2 actors; 7 emotions	Emotion label (angry, disgust, fear, happy, neutral, pleasant surprise, sad), word, speaker ID
emotion2vec pre-training data	Unlabeled emotional speech for self-supervised pre-training of emotion2vec.	Model weights via FunASR	40,000+ hours	Raw audio waveforms; multilingual; diverse environments
Session-level stress data	Stress scores and session feature vectors generated during operation.	Generated by pipeline	7-dim vector per session	Pre/post acoustic scores, pre-HRV score, deltas, agreement status

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Dataset / Source	1	2	3	4
WESAD	Highly suitable because ECG allows accurate RR interval extraction [2].	Highly suitable because RR intervals and timestamps support HRV and Cosinor feature extraction [3].	Highly suitable for training four-class stress models.	Highly suitable for Leave-One-Subject-Out Cross-Validation.
SWELL HRV	Suitable because it provides HRV-related data [23]	Partially suitable because only common HRV features can be aligned with WESAD.	Suitable for auxiliary XGBoost and transfer learning experiments.	Suitable for testing cross-dataset generalization.
MAX30102 PPG stream	Suitable for future wearable-based IBI extraction.	Suitable after converting raw PPG to IBI and HRV features.	Not used for training yet, but suitable for prototype inference.	Useful for real-time prototype validation.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

Data Source	Obj. 1	Obj. 2	Obj. 3	Obj. 4
User preference data	Highly relevant	Highly relevant	Moderately relevant	Relevant
Stress classification data	Highly relevant	Highly relevant	Highly relevant	Highly relevant
VR environment state logs	Moderately relevant	Highly relevant	Highly relevant	Highly relevant

RL interaction logs	Not directly relevant	Moderately relevant	Highly relevant	Highly relevant
Synthetic user profiles	Moderately relevant	Relevant	Highly relevant	Highly relevant
Session feedback data	Relevant	Relevant	Relevant	Highly relevant

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

The component uses a combination of designed preference data and generated synthetic data because real physiological sessions are not yet available at scale. Suitability is therefore assessed against each objective in terms of direct usefulness to formulation, simulation, controller comparison, and runtime validation.

Data source	Obj. 1 (Profile mapping)	Obj.2 (Simulation)	Obj.3 (Controller evaluation)	Obj.4 (Runtime validation)
Long-term audio preference schema	Highly relevant	Moderately relevant	Highly relevant	Moderately relevant
Synthetic user cohort	Highly relevant	Highly relevant	Highly relevant	Moderately relevant
Simulated RL transition records	Not directly relevant	Highly relevant	Highly relevant	Relevant
Unity prototype runtime logs	Not directly relevant	Moderately relevant	Relevant	Highly relevant

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Data Source	Obj. 1 (Ambient)	Obj. 2 (Stress Scoring)	Obj. 3 (Comparison)	Obj. 4 (Anomaly)
RAVDESS	Not directly relevant	Highly relevant	Highly relevant	Moderately relevant

CREMA-D	Not directly relevant	Highly relevant	Highly relevant	Moderately relevant
TESS	Not directly relevant	Highly relevant	Highly relevant	Moderately relevant
emotion2vec pre-training	Not directly relevant	Highly relevant	Highly relevant	Not directly relevant
Session-level data	Not directly relevant	Moderately relevant	Relevant	Highly relevant

3. Methodology

3.1. Data Preprocessing:

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Objective 1 - To extract and preprocess ECG/PPG-based RR/IBI data for HRV stress analysis.

	Transformation Techniques				
Data Source	**Data Cleaning**	**Data Type Conversion**	**Outlier Detection and Handling**	**Data Mapping**	**Data Normalization**
WESAD	ECG signal was cleaned before R-peak detection using NeuroKit2.	R-peak locations were converted into RR intervals in milliseconds.	RR intervals below 300 ms and above 2000 ms were removed. Sudden beat-to-beat changes were filtered and short gaps were interpolated.	WESAD labels were mapped into four classes: relaxed, mild stress, moderate stress, and high stress.	RR intervals were prepared for window-based HRV feature extraction.
SWELL HRV	Invalid or irrelevant records	HRV feature columns were converted	Missing or incompatible records were	SWELL stress conditions	HRV features were prepared for common

	such as rest-condition records were removed where required.	into numerical format for analysis.	removed before feature alignment.	were mapped into available stress categories such as no stress, interruption, and time pressure.	feature alignment with WESAD.
MAX30102 PPG stream	Raw IR PPG signal will be cleaned to reduce motion artifacts and sensor noise.	Pulse peaks will be converted into IBI values in milliseconds.	Invalid pulse intervals and noisy IBI values will be removed before HRV extraction.	The sensor does not provide ground-truth labels; predicted stress labels will be generated by the model.	IBI values will be normalized before real-time inference.

Objective 2 - To apply Cosinor rhythmometry for estimating personalized physiological baselines.

	Transformation Techniques				
Data Source	**Date/Time Transformation**	**Feature Engineering**	**Model-Based Transformation**	**Parameter Estimation**	**Data Aggregation**
WESAD	Beat timestamps were converted into time-of-day values.	Time-aware rhythm features were created from RR/HRV windows.	Subject-wise Cosinor rhythm modelling was applied to estimate expected rhythm baseline.	MESOR, amplitude, and acrophase were estimated for each subject.	Beat-level rhythm information was aggregated into subject-wise rhythm parameters.
SWELL HRV	Timestamp-based processing was applied where sufficient time information was available.	HRV features were prepared for possible rhythm-aware analysis.	Cosinor modelling was limited due to dataset differences and timestamp availability.	Rhythm parameters were estimated only where sufficient longitudinal data was available.	Dataset-level HRV information was aligned for auxiliary analysis.

MAX30102 PPG stream	Timestamped IBI values will be converted into time-of-day values.	Longitudinal IBI/HRV patterns will be used for future rhythm modelling.	Cosinor fitting can be updated when sufficient user-specific wearable data is collected.	User-specific MESOR, amplitude, and acrophase can be estimated after calibration.	Real-time IBI records can be accumulated to update personalized baseline information.

Objective 3 - To develop HRV, circadian, and Cosinor residual features for stress inference.

	Transformation Techniques				
Data Source	**Feature Engineering**	**Date/Time Transformation**	**Data Integration**	**Data Scaling**	**Data Standardization**
WESAD	HRV features such as mean RR, SDNN, RMSSD, pNN50, LF, HF, LF/HF, SD1, and SD2 were extracted. Cosinor residual features such as residual mean, residual standard deviation, residual maximum, residual trend, and residual energy were also extracted.	Circadian features were generated using time-of-day sine and cosine transformations.	HRV features, Cosinor residuals, Cosinor parameters, and circadian features were combined into one feature vector.	Numerical feature values were scaled before model training.	Structured feature matrices were standardized for machine learning models.
SWELL HRV	Common HRV features such as mean RR,	Limited time-aware transformation was possible	SWELL HRV features were	Common HRV feature values	Feature names and formats were standardized to match WESAD-

	SDNN, RMSSD, LF, HF, LF/HF, SD1, and SD2 were selected.	depending on available timestamp information.	aligned with WESAD common features for cross-dataset preparation.	were scaled before training.	compatible inputs.
MAX30102 PPG stream	IBI values will be converted into HRV features. Cosinor residuals can be calculated after a baseline is available.	Timestamped IBI values will support time-of-day feature generation.	Real-time HRV and residual features will be passed into the stress inference pipeline.	Real-time feature scaling will be applied before prediction .	Feature format will be standardized to match the trained model input format.

Objective 4 - To train and evaluate ML/DL models for four-level stress classification.

	Transformation Techniques				
Data Source	**Data Scaling**	**Data Normalization**	**Feature Selection**	**Data Formatting**	**Label Encoding**
WESAD	HRV, circadian, and Cosinor-based features were scaled before classification.	RR and Cosinor residual sequences were normalized for sequence-based model input.	Standard HRV features, circadian features, Cosinor residual features, and Cosinor parameters were selected.	Data was prepared as both structured feature matrices and sequence windows.	WESAD labels were encoded into four classes: relaxed, mild, moderate, and high stress.
SWELL HRV	Common HRV features were scaled before auxiliary analysis.	HRV feature values were normalized for compatibility with WESAD-based inputs.	Only HRV features common to both WESAD and SWELL were selected.	SWELL features were formatted to match WESAD common-feature structure.	SWELL conditions were encoded into available stress categories.

MAX30102 PPG stream	Real-time IBI-derived features will be scaled before inference.	IBI windows will be normalized before prediction.	PPG-derived IBI, HRV, and future Cosinor residual features will be selected.	Real-time data will be formatted as input windows for the trained pipeline.	No ground-truth label encoding is available from the sensor; predicted labels will be produced by the model.

Objective 5 - To assess robustness, generalization, and real-time applicability of the stress inference engine.

	Transformation Techniques			
Data Source	**Data Partitioning**	**Data Integration**	**Performance Evaluation Preparation**	**Real-Time Transformation**
WESAD	Subject-wise partitions were prepared for Leave-One-Subject-Out Cross-Validation.	WESAD feature matrices and labels were combined for model evaluation.	Accuracy, macro F1-score, kappa, and classification reports were prepared.	Offline benchmark data was used before real-time deployment.
SWELL HRV	SWELL records were separated and prepared for auxiliary analysis.	Common HRV features were integrated with WESAD-compatible features.	Cross-dataset evaluation inputs were prepared.	SWELL was used to test robustness across different stress conditions.
MAX30102 PPG stream	Real-time windows will be formed from incoming IBI streams.	PPG-derived IBI features will be integrated into the trained inference pipeline.	Future latency and throughput evaluation data will be prepared.	Raw PPG will be transformed into IBI, HRV features, and model input windows.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

The preprocessing stage converts user preferences, stress classes, VR parameters, and session logs into a numerical form suitable for RL training and evaluation. Since the agent uses a structured observation vector, all categorical values are encoded, continuous values are normalized, missing or invalid records are handled, and safety constraints are applied to keep VR parameters within comfortable ranges

Objective 1 - Prepare user preference and stress classification inputs

Data Source	Transformation Technique	Purpose
User preference data	Categorical encoding	Convert preferences such as color tone, scene density, and visual comfort level into numeric values.
User preference data	Normalization	Scale intensity-based preference values such as brightness preference and motion tolerance to a 0-1 range.
Stress classification data	Label encoding	Map low, moderate, high, and very high stress classes into numeric stress levels.
Stress classification data	Timestamp alignment	Align stress outputs with the corresponding VR session and timestep.

Objective 2 - Rule-based initialization preprocessing

Data Source	Transformation Technique	Purpose
Preference + stress data	Data mapping	Map stress level and user preferences to initial VR parameters using predefined rules.
VR parameter ranges	Safety clipping	Restrict brightness, fog, density, and motion values to safe comfort boundaries.
Synthetic profiles	Profile construction	Create simulated user profiles with known preference and response patterns for early testing.

Objective 3 - RL interaction preprocessing

Data Source	Transformation Technique	Purpose
RL observations	Feature vector construction	Combine stress level, preference vector, VR state, previous action, and session progress into the observation vector.
RL actions	Action scaling	Convert the agent's action values into valid parameter adjustments for the Unity environment.
Reward values	Reward calculation	Calculating rewards using stress reduction, stress trend, smoothness, comfort, and terminal relaxation bonus.

Objective 4 - Evaluation preprocessing

Data Source	Transformation Technique	Purpose
Session logs	Outlier handling	Remove invalid records from interrupted sessions, missing stress classifications, or impossible parameter values.
Reward logs	Aggregation	Compute average reward per episode, reward trend, and final session reward.
Feedback data	Score conversion	Convert comfort and relaxation feedback into numerical evaluation indicators.

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

Objective 1 — Preference profile construction preprocessing

Data Source	Transformation Technique	Purpose
Long-term preference fields	Categorical normalization of all vocabulary terms	Ensure consistent labels such as slow/medium/fast or low/medium/high across the schema
Preferred instrument set	Multi-hot encoding of selectable instruments	Represent multiple simultaneous instrument preferences
Ordinal preference fields	Monotonic mapping to [0,1] target anchors	Convert tempo, intensity, brightness, novelty, rhythm, reverb, and volume into

		controller-ready values
Ambient vs music balance	Complementary target generation (ambient mix + music mix = 1)	Maintain physically valid audio mixing targets
Sound style	One-hot identity encoding and style-to-scene mapping	Preserve semantic ambience such as forest, ocean, rain, temple, or studio
Synthetic user records	Tolerance-width sampling	Model how strictly each user reacts to deviations from preferred settings

Objective 2 — Simulated-user environment preprocessing

Data Source	Transformation Technique	Purpose
Synthetic user profiles	User-level train/validation/test split	Support unseen-user evaluation and reduce leakage between simulated users
Continuous targets	Parameter clipping to [0,1]	Keep control parameters within safe and interpretable bounds
Stress and confidence proxies	Min-max normalization and bounded simulation rules	Standardize the state representation used by the controller
State history	Rolling-window aggregation	Provide recent trend information for stress and confidence
Action deltas	Bounded scaling per step	Prevent unsafe large jumps in audio parameters
Recent change counts	Discrete-to-continuous normalization	Track novelty exposure and intervention frequency

Objective 3 — Controller comparison and policy-ready preprocessing

Data Source	Transformation Technique	Purpose
State variables	Feature stacking / concatenation	Create a single controller state vector from preferences, current parameters, and dynamics
Safety flags	Binary encoding	Represent fallback, mute, or low-confidence damping conditions explicitly
Reward components	Weighted multi-objective	Combine preference satisfaction, smoothness, and safety terms

	aggregation	into a controller reward
Controller conditions	Ablation-style tagging	Separate generic static, personalized static, rule-based, and PPO-based conditions for evaluation
Episode logs	Seed-aware partitioning	Support repeated experiments without mixed-seed contamination
Output parameters	Time-aligned action tracing	Measure intervention frequency and adaptation smoothness

Objective 4 — Unity runtime validation preprocessing

Data Source	Transformation Technique	Purpose
Policy outputs	Inference-friendly serialization	Prepare selected policy outputs for Unity-side playback control
Runtime traces	Timestamp alignment and smoothing	Enable clear visualization of input and output changes over time
Session logs	CSV export	Support later offline analysis and dashboard plotting
Dashboard fields	Derived metric calculation	Compute intervention frequency, smoothness, and safety-mode proportions
Safety states	Human-readable label mapping	Make controller behavior interpretable during presentation and debugging

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Objective 1 — Ambient environment assessment preprocessing

Data Source	Transformation Technique	Purpose
Raw ambient audio	Resampling to 16kHz	Standardize sample rate for consistent acoustic analysis
Raw ambient audio	Mono conversion	Convert stereo recordings to single-channel audio
Raw ambient audio	Float32 normalization	Normalize amplitude values to [-1, 1] range for consistent RMS computation

Objective 2 — Voice-based stress scoring preprocessing

Data Source	Transformation Technique	Purpose
RAVDESS / CREMA-D / TESS	Emotion-to-stress label mapping	Map 8 emotion categories to binary stress labels for ablation study
RAVDESS / CREMA-D / TESS	Speaker-independent split (80/10/10)	Ensure test speakers are unseen during training to prevent data leakage
Combined dataset	Calm class augmentation (192 → 1,920)	Address class imbalance using pitch shifting, time stretching, and noise
Raw audio waveforms	Resampling to 16kHz	Match input requirements for wav2vec2 and emotion2vec encoders
Long recordings (>10s)	Chunking into 6-second segments	Split long recordings into segments for emotion2vec; skip silent chunks
emotion2vec output	Emotion probability → stress mapping	Map 9 emotion probabilities to 0–10 stress via valence-arousal weights

Objective 3 — Pre/post comparison and cross-modal validation preprocessing

Data Source	Transformation Technique	Purpose
Pre/post stress scores	Absolute and relative difference	Calculate stress change magnitude and direction
Voice + HRV scores	Threshold-based agreement	Classify modality agreement (within 2.5 points) or divergence

Objective 4 — Autoencoder anomaly detection preprocessing

Data Source	Transformation Technique	Purpose
Session feature vectors	Min-max normalization [0, 1]	Scale all 7 features to uniform range for autoencoder training
Session records	Normal session filtering	Select sessions with meaningful stress reduction for training
Session records	Outlier removal	Remove sessions with invalid or missing scores

3.2. Scalability

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

WESAD dataset can be effectively applied for initial stress inference model building and benchmark testing since it comprises labelled physiological data and is commonly used for research on wearable stress detectors. Still, the small number of valid subjects in the dataset restricts the possibility of reliable model testing with straightforward train/test splitting. For that reason, LOOCV was used to make it possible for each of the participants to be validated as an unseen subject, which gives a more reliable indication of how the system would work with unknown users.

SWELL HRV dataset makes the analysis scalable through the addition of a supplementary dataset captured under different stress scenarios. The use of cross-dataset analysis enables validation of whether the patterns of heart rate variability associated with different levels of stress revealed in WESAD dataset generalize to another dataset. Nonetheless, SWELL HRV uses a different labeling and cannot be directly aligned with four stress levels of this component. Hence, only those HRV features that are common both for WESAD and SWELL HRV datasets were selected for the purpose of cross-dataset analysis.

MAX30102 PPG Stream contributes towards the scalability of the proposed solution by acting as a future wearable input source for the prototype. While the WESAD database is used mostly for offline training and validation, MAX30102 offers a live PPG source from a wearable device. The data from this sensor can easily be translated into pulse peaks, IBI, HRV features, and finally, Cosinor residuals. Since both IBI and RR intervals represent beat-to-beat intervals, the same approach of extracting HRV and Cosinor features followed by stress prediction can be deployed with appropriate processing and artifact removal.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

The data collection process is scalable because every VR session automatically generates interaction data. Unlike a fixed dataset, the amount of usable training and evaluation data increases as more users complete relaxation sessions. For example, a 14-minute active RL phase with updates every 10 seconds can generate approximately 84 state-action-reward records per session. With 30 pilot sessions, the system can produce approximately 2,520 transition records, excluding additional warm-up and feedback data.

The pipeline can also be scaled from synthetic testing to real-world deployment. Initially, synthetic profiles validate agent logic and safety constraints. Later, real pilot-session logs can be appended to improve personalization and evaluate whether the agent adapts to different user preference patterns

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

The preference-driven design is inherently scalable. The long-term schema already defines 66.1 million theoretical profile combinations when preferred instruments are treated as any non-empty subset of six selectable instruments. This means the simulation environment can generate large user cohorts without changing the controller architecture. The state and action spaces are also compact enough for continuous-control algorithms, because the controller operates on a limited set of interpretable audio dimensions rather than raw audio samples. As the project matures, the same representation can be expanded with session-level preferences and real stress/confidence inputs from Components A and B without redesigning the entire pipeline

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

The data pipeline is designed to scale across three dimensions. First, the acted speech dataset (9,227 samples) is sufficient for ablation study evaluation and can be expanded by adding additional public datasets without architectural changes. Second, the finalized emotion2vec_base_finetuned model was pre-trained on over 40,000 hours of emotional speech, providing a large-scale foundation that does not require retraining for new speakers or environments. Third, the autoencoder anomaly detection model improves with scale — as more real user sessions are collected, the normal pattern representation becomes more robust. The chunked scoring approach (6-second segments averaged) allows the system to process recordings of any duration without requiring fixed-length inputs.

3.3. Feature extraction

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Feature Group	Features Extracted	Extraction Method	Purpose
Time-domain HRV features	Mean RR, SDNN, RMSSD, pNN50, CV_RR	Calculated directly from RR/IBI windows.	Measures overall and short-term heart rhythm variability.
Frequency-domain HRV features	VLF, LF, HF, LF/HF, LF_NU	RR intervals were resampled and power spectral density was estimated.	Represents autonomic nervous system activity patterns.

Non-linear HRV features	SD1, SD2, SD ratio	Derived from Poincaré-style RR interval analysis.	Captures short-term and long-term variability structure.
Circadian features	sin_24h, cos_24h, sin_90m, cos_90m, cortisol proxy	Calculated from the midpoint timestamp of each window.	Encodes time-of-day and rhythm context.
Cosinor parameters	MESOR, amplitude, acrophase	Estimated through subject-wise Cosinor modelling.	Represents personalized rhythm baseline characteristics.
Cosinor residual features	res_mean, res_std, res_max, res_trend, res_energy	Calculated from actual RR minus expected Cosinor baseline.	Captures deviation from expected rhythm baseline.
Model 7 sequence channels	RR_norm, RMSSD_roll, SDNN_roll, HR, Cosinor_residual	Calculated for each beat in a 120-beat window.	Allows CNN-BiLSTM-Attention to learn temporal stress patterns.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

Feature Group	Extracted Features
Stress features	Current stress class, previous stress class, stress trend, stress stability over recent timesteps
Preference features	Preferred brightness, color tone, fog preference, scene density preference, visual motion sensitivity
VR environment features	Brightness level, fog density, sky tone, forest object density, animation/motion intensity, breathing guide visibility
Session features	Elapsed time, session phase, timestep number, number of changes already applied
RL action features	Selected parameter adjustment, action magnitude, previous action, smoothness of change
Reward features	Stress reduction reward, stress trend reward, smoothness penalty, comfort penalty, terminal low-stress bonus

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

Feature Group	Extracted Features
Preference encodings	One-hot sound style, multi-hot instrument vector, ordinal encodings for tempo, intensity, brightness, novelty, rhythm, reverb, volume

Continuous target audio profile	Target intensity, brightness, tempo, ambient mix, music mix, novelty amount, reverb amount, rhythm amount, dissonance allowance, volume level
User sensitivity features	Tolerance widths for novelty, brightness, tempo, rhythm, and volume deviations
Dynamic controller state	Stress proxy, confidence, stress trend, confidence trend, recent action history, time since session start, recent change count
Action features	Delta intensity, delta density, delta brightness, delta tempo, delta fade, delta music mix, delta ambient mix
Evaluation features	Preference satisfaction, smoothness score, safety violation count, intervention frequency, relaxation/calm proxy

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Feature Group	Extracted Features
Ambient acoustic features (Layer 1)	Root Mean Square (RMS) energy, Zero Crossing Rate (ZCR) — computed from raw audio using librosa
Learned representations (Layer 2 — ablation)	768-dimensional embedding vectors from wav2vec2-base and emotion2vec_base encoders; captures pitch variation, vocal tension, speech rhythm, spectral energy
Emotion classification (Layer 2 — finalized)	9-class emotion probability distribution from emotion2vec_base_finetuned: angry, fearful, disgusted, sad, surprised, neutral, happy, other, unknown
Stress score (Layer 2)	Continuous 0–10 score = weighted sum of emotion probabilities × valence-arousal stress weights
Session comparison (Layer 3)	Pre-score, post-score, absolute change, relative change, outcome, magnitude
Cross-modal (Layer 4)	Voice pre-score, HRV pre-score, difference, agreement status, confidence level
Anomaly detection (Layer 5)	7-dim vector: pre-acoustic, post-acoustic, pre-HRV, acoustic delta, HRV delta, cross-modal agreement, duration

4. Modelling and Results

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

This section describes the modelling and result analysis process that produced the most important findings following the creation of HRV, circadian and Cosinor features. This section aims to determine if rhythmic feature extraction leads to improved classification of stress across multiple levels. Evaluation was done according to the outline provided in the proposal, with model comparison, ablation study and cross-dataset evaluation among others.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

The modelling approach is based on Reinforcement Learning rather than supervised classification or regression. The adaptive VR environment is treated as an environment in which the agent observes the current user/context state, selects an action that changes VR parameters, receives a reward, and updates its policy. The planned algorithm is Proximal Policy Optimization (PPO), wrapped with Context-Aware Model-Agnostic Meta-Learning (CA-MAML) to improve adaptation across users with different relaxation preferences. The rule-based initializer is used before the active RL phase to provide a safe and suitable starting state.
The current stage focuses on validating the data structure and evaluation pipeline using synthetic profiles and simulated interaction logs. Final empirical results will be completed after real pilot sessions. Therefore, the charts below are presented as illustrative analysis outputs that show how the collected/generated data will be analyzed

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

The current prototype stage does not yet include a completed real-user physiological evaluation. Accordingly, the modelling section emphasizes the structure of the simulation environment, the mapping from preferences to controller targets, and the evaluation conditions that will later be used to validate the selected adaptive policy. Where quantitative values are reported below, they are drawn from the synthetic-user cohort and the designed preference space used for prototype analysis

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

The modelling approach progressed through three stages as an ablation study, evaluating different encoder-classifier combinations to identify the most effective approach for voice-based stress detection. Stage 1 used wav2vec2-base encoder with a binary MLP classifier as the baseline. Stage 2 switched to emotion2vec_base encoder with the same binary MLP. Stage 3 (finalized) used emotion2vec_base_finetuned for direct emotion classification with valence-arousal stress mapping. The autoencoder anomaly detection model was trained separately with architecture
7→16→8→4→8→16→7 using MSE loss.

4.1. Key Insights:

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Seven model configurations were evaluated using Leave-One-Subject-Out Cross-Validation (LOSO-CV). This validation method was used because the system is expected to work for unseen users. Therefore, each subject was tested separately while the remaining subjects were used for training. The final implementation stores the results for seven models and prints the final comparison using accuracy, macro F1-score, and quadratic weighted kappa.

No.	Model	Accuracy	Macro F1	Kappa	Format
1	XGBoost HRV Only	0.773	0.549	0.599	.pkl
2	XGBoost HRV + Cosinor	0.842	0.628	0.769	.pkl
3	XGBoost SWELL + WESAD	0.675	0.460	0.437	.pkl
4	CNN-LSTM Circadian	0.695	0.450	0.486	.keras
5	CNN-LSTM Transfer Learning	0.524	0.358	0.336	.keras
6	Integrated Cosinor-CNN-LSTM	0.663	0.423	0.402	.keras
7	CNN-BiLSTM-Attention + Cosinor	0.794	0.522	0.683	.keras

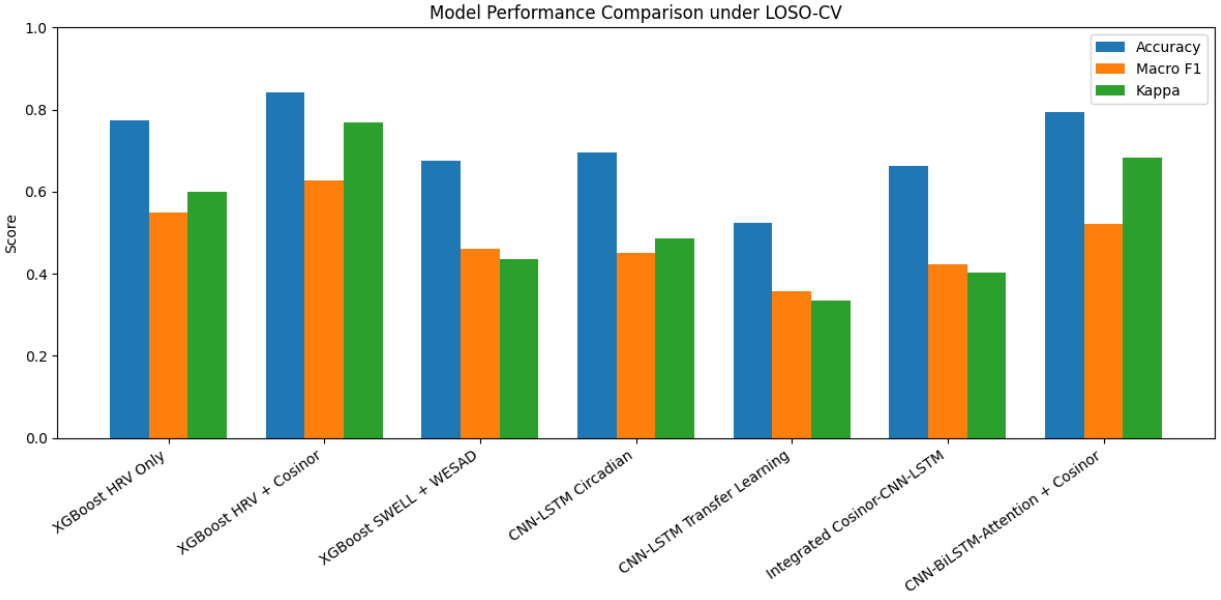


Figure 1:Model performance comparison using Accuracy, Macro F1-score, and Kappa under Leave-One-Subject-Out Cross-Validation.

Figure 1 shows the comparison of Accuracy, Macro F1-score, and Kappa across all seven models. The XGBoost HRV + Cosinor model achieved the highest accuracy and macro F1-score, while the CNN-BiLSTM-Attention + Cosinor model also showed strong agreement based on kappa.

The outcomes indicate that the best overall model was the XGBoost HRV + Cosinor model, which had an accuracy score of 0.842, macro F1-score of 0.628, and kappa of 0.769. Compared to the XGBoost HRV-only model, adding Cosinor features resulted in an improvement of accuracy from 0.773 to 0.842, macro F1-score from 0.549 to 0.628, and kappa from 0.599 to 0.769. This suggests that using rhythm-aware residual features assisted the model in recognizing changes associated with stress as opposed to natural variations in the physiological rhythm.

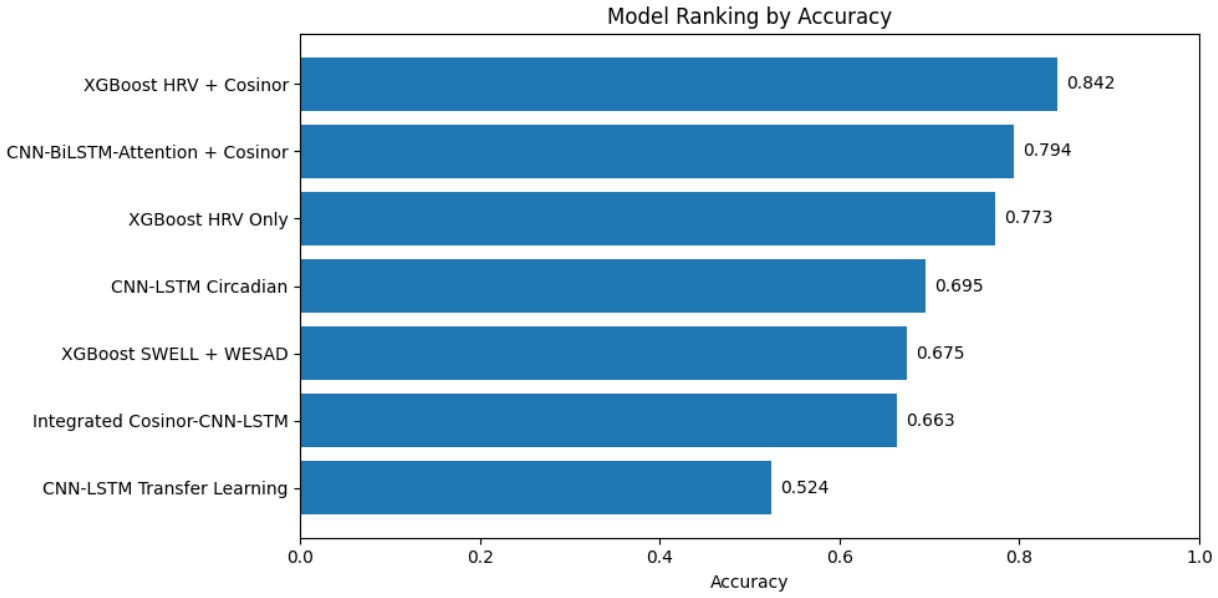


Figure 2: Ablation comparison between XGBoost HRV Only and XGBoost HRV + Cosinor. The Cosinor-enhanced model improved all three evaluation metrics.

Figure 2 presents the ablation comparison between the HRV-only model and the HRV + Cosinor model. The improvement across all three metrics indicates that rhythm-aware residual features helped the model recognize stress-related changes more effectively than using HRV features alone.

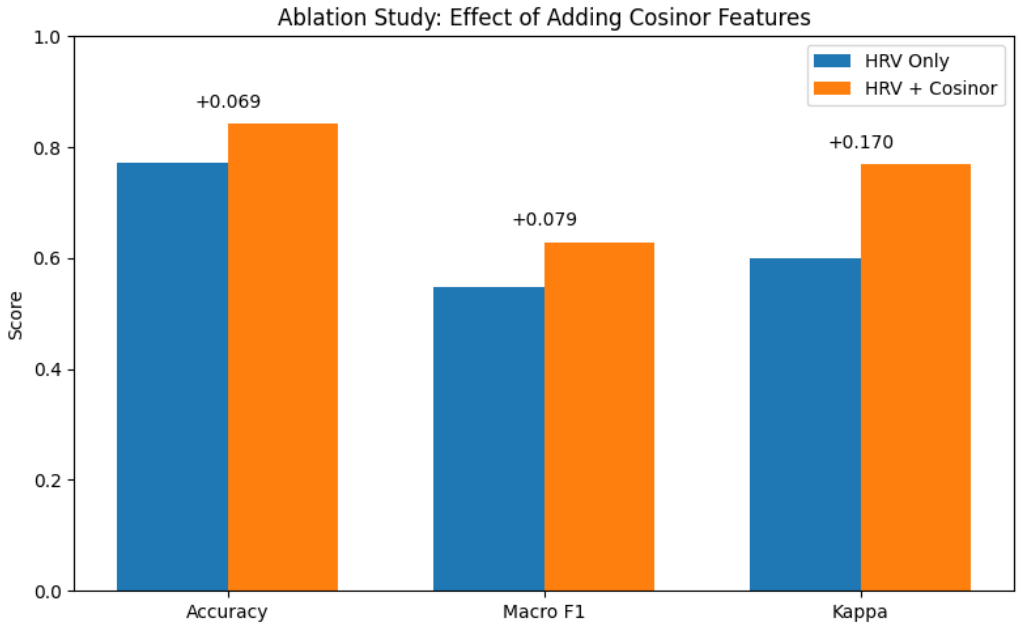


Figure 3: Accuracy-based ranking of all seven evaluated models. XGBoost HRV + Cosinor achieved the highest accuracy

Figure 3 ranks all models according to accuracy. The XGBoost HRV + Cosinor model achieved the highest accuracy, followed by the CNN-BiLSTM-Attention + Cosinor model and the XGBoost HRV-only model.

The key takeaway from this comparison is that Cosinor-based rhythm-aware feature engineering enhanced the performance of the structured HRV model. According to the proposal, it is possible to classify natural variations in HRV rhythms as stress using existing detection models, which fail to account for temporal biological contexts. The improved performance of the HRV + Cosinor model supports this research direction because the model used both standard HRV features and deviations from an expected rhythm baseline

Also, the CNN-BiLSTM-Attention + Cosinor achieved high value of Kappa coefficient with 0.683, which demonstrates the potential of temporal deep learning in detecting stress using sequential data. It utilized five-channel sequence including normalized RR, rolling RMSSD, rolling SDNN, heart rate, and Cosinor residual, as well as seven circadian/rhythm features. According to the metadata on its implementation, this model follows Conv1D × 2 → BiLSTM → Self-Attention → Circadian fusion → Softmax architecture and takes two input arrays of shape (120, 5) and (7,) .

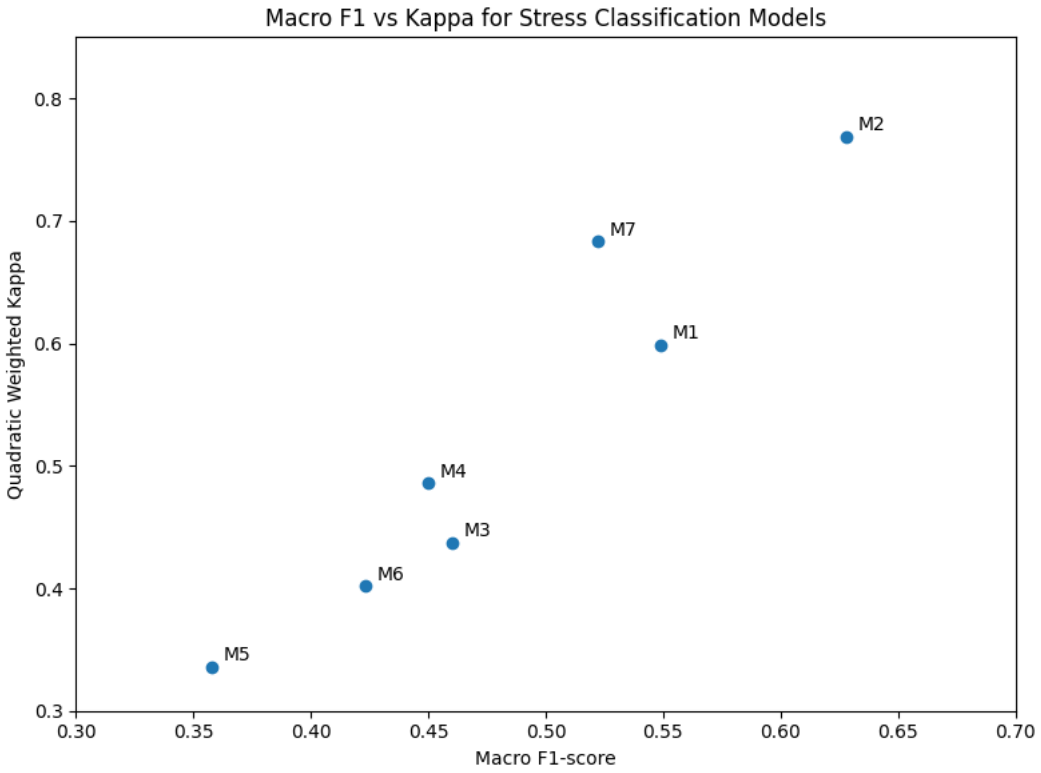


Figure 4: Relationship between Macro F1-score and Kappa. Models closer to the upper-right region show stronger balanced classification performance and agreement.

Figure 4 shows the relationship between Macro F1-score and Kappa. Models positioned toward the upper-right side show stronger balanced classification performance and better agreement. The XGBoost HRV + Cosinor model appears as the strongest overall model, while CNN-BiLSTM-Attention + Cosinor shows competitive agreement performance.

SWELL XGBoost model in combination with the WESAD dataset showed worse results compared to the WESAD-only Cosinor model. This is logical since SWELL has other labels and only similar HRV features could be used for alignment purposes. Consequently, SWELL was valuable more for checking the robustness between datasets than being an alternative model itself.

Similarly, the CNN-LSTM Transfer Learning approach had poorer-than-expected performance. This might be caused by the mismatch of domains between SWELL and WESAD, differing stress labels, and the lack of a high stress level category in SWELL. Thus, while transfer learning was explored, its findings imply that the domain adaptation necessary for transfer learning is still needed for SWELL and WESAD.

In conclusion, the key outcome to note is that incorporating Cosinor rhythm-aware features enhanced the HRV baseline model. As such, this contributes to the core innovation of the component: the use of stress detection through comparison of present HRV rhythm to physiological rhythm rather than HRV features alone.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

Initial synthetic analysis indicates that the proposed data sources are sufficient to support the RL pipeline. The stress classification input provides the main user-state signal, preference data provides personalization constraints, VR state logs describe the controllable environment, and RL interaction logs provide the state-action-reward transitions required for agent learning.

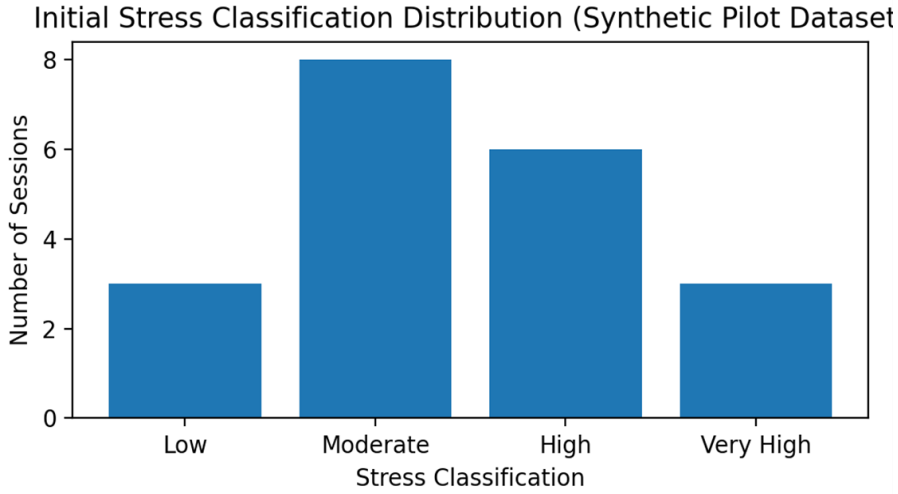


Figure 5: Initial stress classification distribution from a synthetic pilot dataset

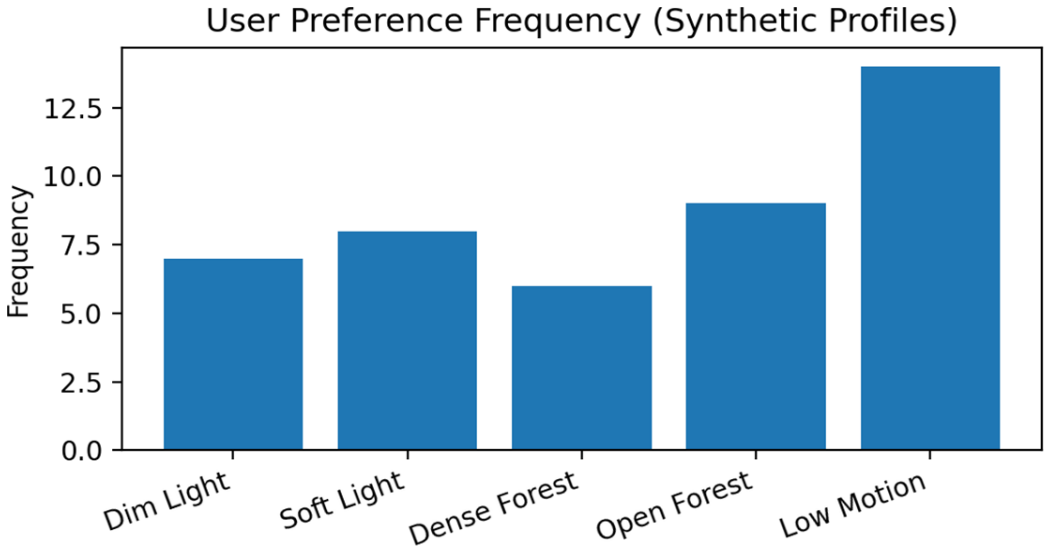


Figure 6: User preference frequency from synthetic user profiles

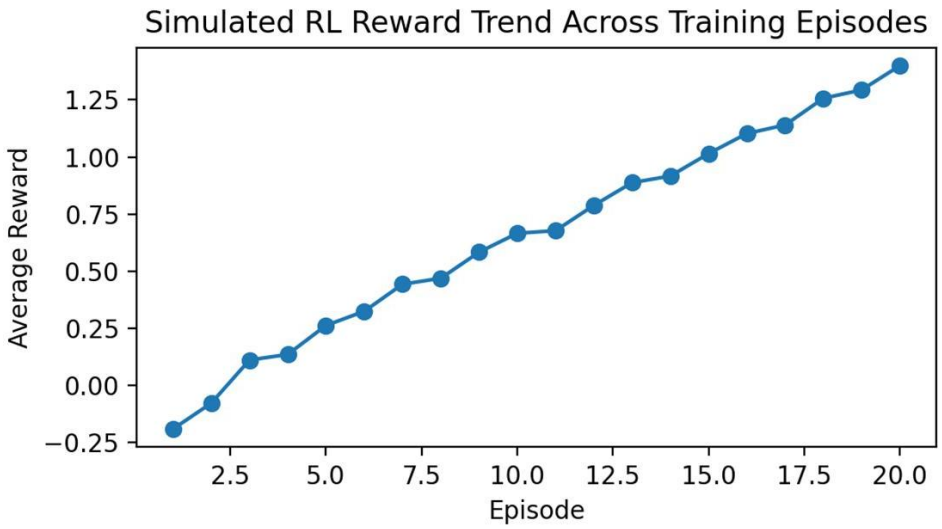


Figure 7: Simulated average reward trend across RL training episodes

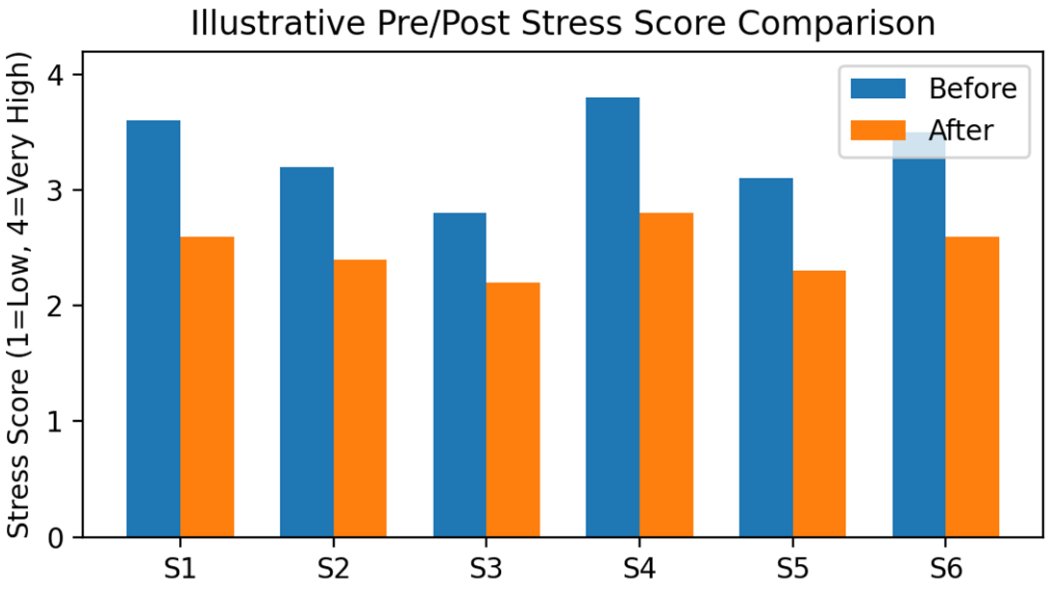


Figure 8: Illustrative before/after stress score comparison for pilot-style sessions

Figure 5 shows how initial stress levels can be analyzed before VR exposure. This helps

identify whether the system is being tested with enough moderate, high, and very high stress cases. Figure 6 shows that user preferences are not uniform; some users prefer lower motion while others prefer brighter or more open environments. This supports the need for personalized adaptation rather than a fixed VR scene.

Figure 7 shows the expected pattern of RL learning, where average reward improves across episodes as the agent learns safer and more effective parameter adjustments. Figure 8 shows the type of before/after stress comparison that can be used during pilot evaluation. In the final evaluation, this analysis will use real stress-classification outputs and user feedback instead of synthetic examples.

The following metrics will be used to evaluate the adaptive component:

Metric	Purpose	Expected direction
Average episode reward	Measures whether the RL policy is improving over repeated sessions	Increase
Stress reduction score	Compares stress level before and after the active VR phase	Decrease in stress
Stress trend reward	Checks whether stress decreases gradually during the active RL phase	Positive trend
Action smoothness penalty	Ensures that environmental changes are not abrupt or uncomfortable	Low penalty
Comfort/discomfort rating	Evaluates nausea, visual discomfort, and overall comfort	High comfort / low discomfort
Preference satisfaction rating	Checks whether the final environment matches the user's relaxation preferences	Increase

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

Preference Space Cardinality Analysis The long-term preference schema is already large enough to justify adaptive personalization. Using five sound styles, six selectable instruments treated as non-empty subsets, and the remaining categorical preference fields, the schema yields 66,134,880 possible unique long-term profiles. This result shows that a single fixed meditation soundtrack cannot reasonably cover the design space addressed by the component.

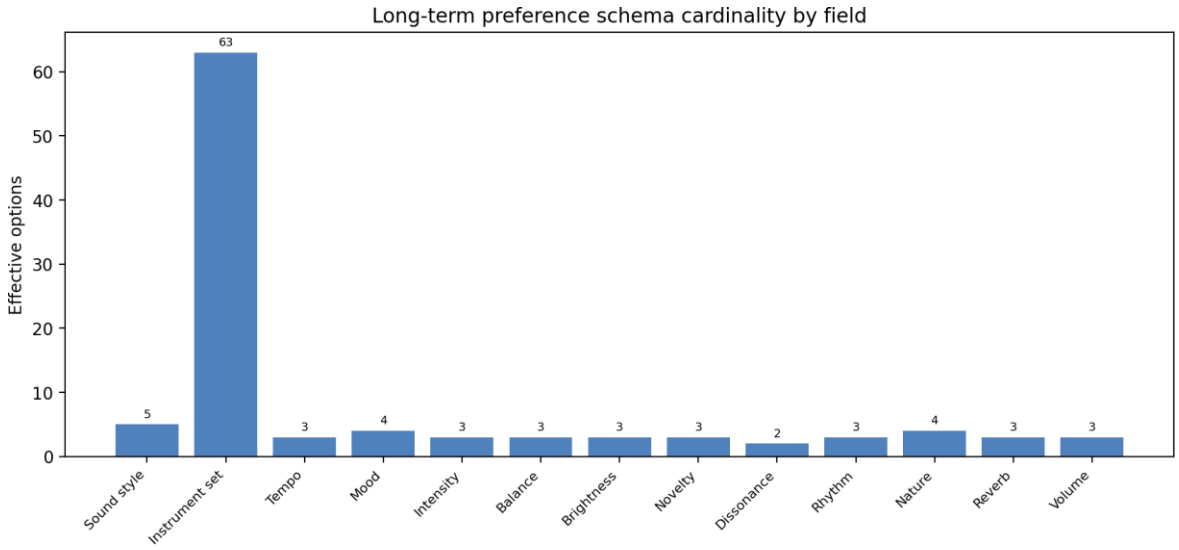


Figure 9: Effective option count by long-term preference field

Continuous Target Distribution Analysis A synthetic cohort of 1,500 users was sampled uniformly across the schema and mapped into continuous controller targets. The resulting target distributions are clearly multimodal rather than uniform, because discrete preferences such as low/medium/high intensity or soft/neutral/bright brightness map to structured anchor regions. This is a useful property for the controller because it produces interpretable continuous targets while preserving the semantics of the original questionnaire.

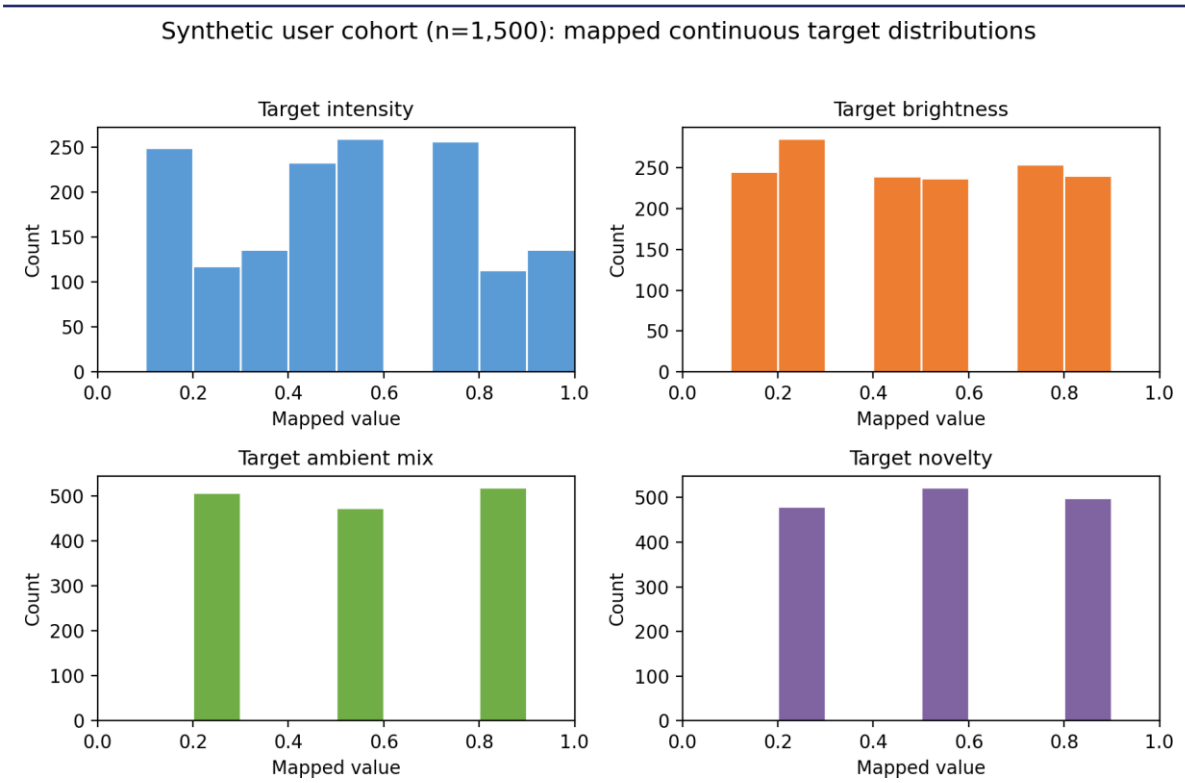


Figure 10: Synthetic user cohort target distributions after categorical-to-continuous mapping

Gap-Focused Evaluation Framework The component's research gap is not satisfied by comparing static music with generated music alone. The more suitable evaluation is an ablation-style comparison that isolates personalization, real-time adaptation, confidence-awareness, and policy learning. Table below, therefore defines four core conditions and one optional condition that can be used to show whether each missing part of the gap contributes meaningfully to the final system.

Condition	Personalized	Real-time Adaptive	Confidence-aware	Primary Purpose
Generic static audio	No	No	No	Lower-bound baseline
Personalized static audio	Yes	No	No	Isolate preference matching without adaptation
Adaptive audio without confidence	Yes	Yes	No	Test whether confidence-awareness matters

Confidence-aware rule-based adaptive audio	Yes	Yes	Yes	Strong safe non-learning baseline
Full PPO-based adaptive audio	Yes	Yes	Yes	Evaluate the complete proposed solution

Selected Prototype Model For the current Unity prototype, PPO is selected as the primary deployable controller because clipped policy updates are compatible with gradual audio adaptation, on-policy rollouts are straightforward to align with simulated sessions, and the policy can be exported and traced cleanly during runtime validation. TD3 and SAC remain valuable comparison models for the later algorithm study, but PPO is the most practical first controller for demonstration because its behavior is easier to visualize and explain in a presentation setting.

Candidate model	Action space fit	Prototype interpretability	Current role
PPO	Strong continuous-control fit	High	Selected primary prototype policy
TD3	Strong continuous-control fit	Medium	Planned comparison baseline
SAC	Strong continuous-control fit	Medium	Planned comparison baseline

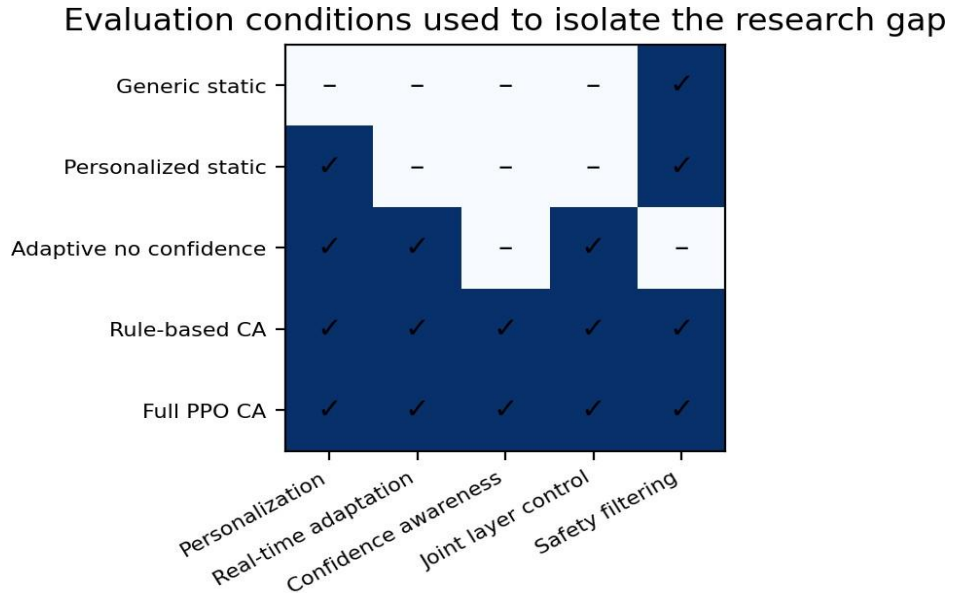


Figure 11: Evaluation conditions used to isolate personalization, adaptation, confidence-awareness, and joint-layer control

Recommended Metrics for the Selected Model The selected PPO model should be judged using metrics that directly match the research gap: preference satisfaction, calmness/relaxation proxy improvement, safety violation rate under low- confidence conditions, smoothness of adaptation, intervention frequency, and stability of ambient–meditation balance. These metrics are more informative for this component than raw reward alone because the goal is safe personalized relaxation support rather than aggressive maximization of a single score.

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Dataset Class Distribution Analysis

The combined dataset of 9,227 samples revealed a significant class imbalance. The calm class — representing the most critical category for post-meditation stress detection — constituted only 2% of the total dataset (192 samples out of 9,227). After mapping to binary stress labels, stressed samples (5,589) outnumbered not-stressed samples (3,638) by a ratio of 1.54:1. Data augmentation was applied to the calm class (192 → 1,920 samples) using pitch shifting (±1–2 semitones), time stretching (0.88x–1.12x), and additive Gaussian noise.

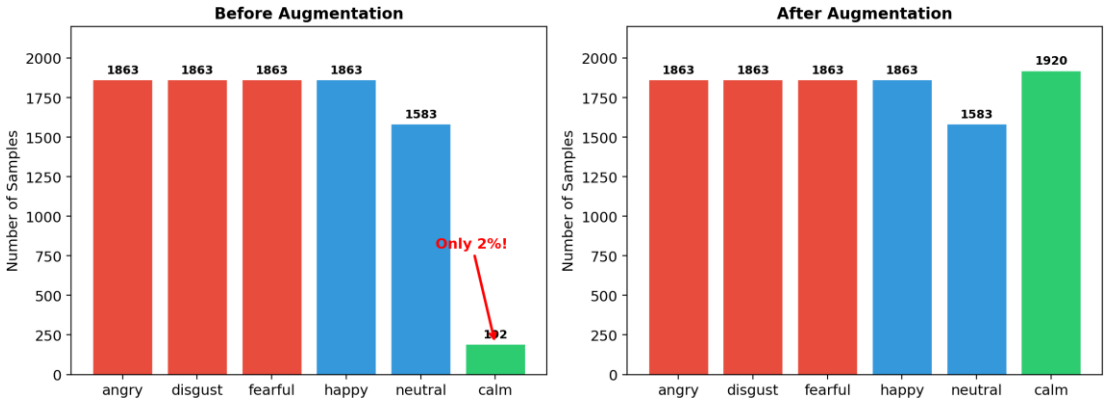


Figure 12: Emotion class distribution before and after augmentation

Ablation Study Results

The three-stage ablation study demonstrated clear performance progression, confirming that emotion-specific pre-training produces significantly better stress representations than speech recognition pre-training.

Approach	Encoder	Method	Test Accuracy	Macro F1
Stage 1	wav2vec2-base (95M, ASR)	Binary MLP	72.54%	0.72
Stage 2	emotion2vec_base (90M, SER)	Binary MLP	84.73%	0.84
Stage 3 (Finalized)	emotion2vec_finetuned (90M, SER)	Emotion → VA mapping	Real-world generalization	N/A

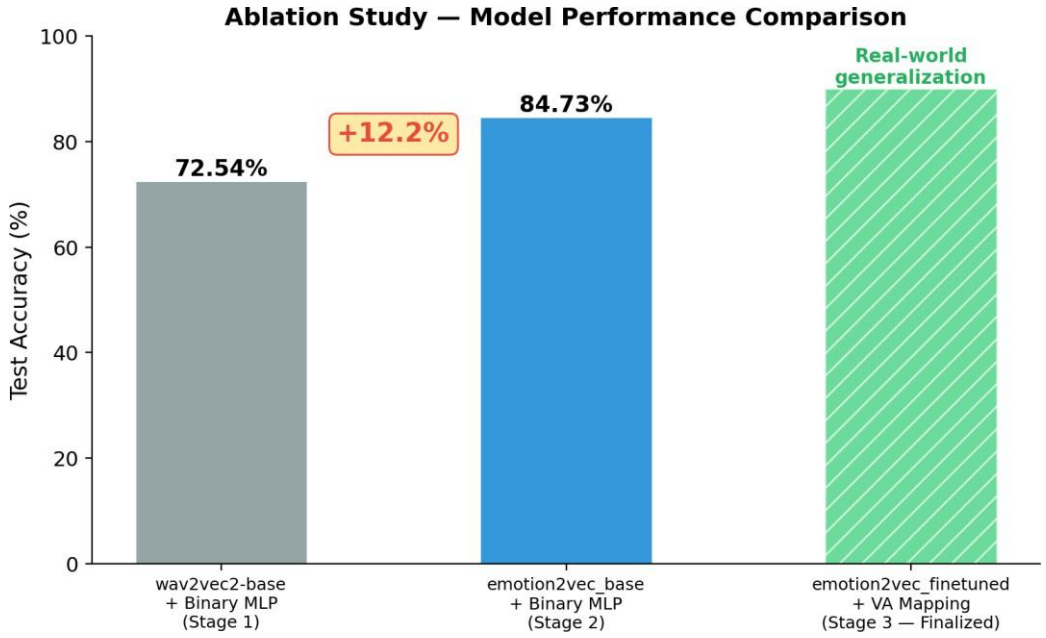


Figure 13: Ablation study — model performance comparison across three stages

Per-Emotion Regression Analysis

The VA regression model (Stage 2) was evaluated per emotion to verify that the valence-arousal mapping correctly differentiates low-stress emotions (calm, neutral, happy) from high-stress emotions (angry, fearful, disgust). Results confirm accurate predictions for all emotions except happy, which is predicted slightly higher due to its high acoustic energy being partially confused with stressed speech.

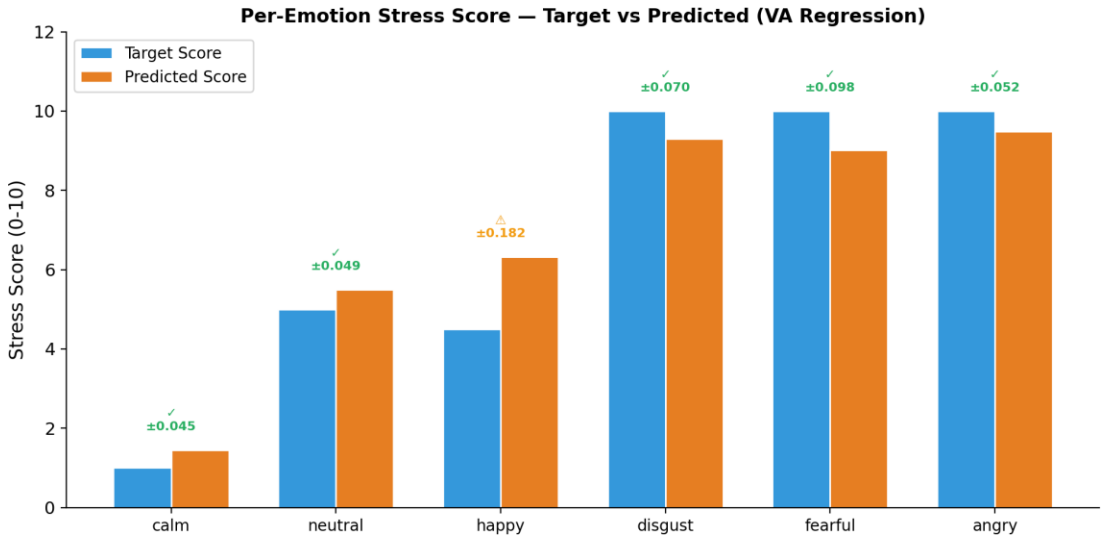


Figure 14: Per-emotion stress score — target vs predicted (VA regression model)

Real-World Voice Testing

The finalized emotion2vec_finetuned approach was tested across three domains: TTS-generated voices with controlled prosody, real human voice recordings, and the RAVDESS acted speech test set. Results confirm correct directional scoring across all domains, with the largest gap on acted speech (8.1 points) and a smaller but correct gap on real human voice (0.87 points).

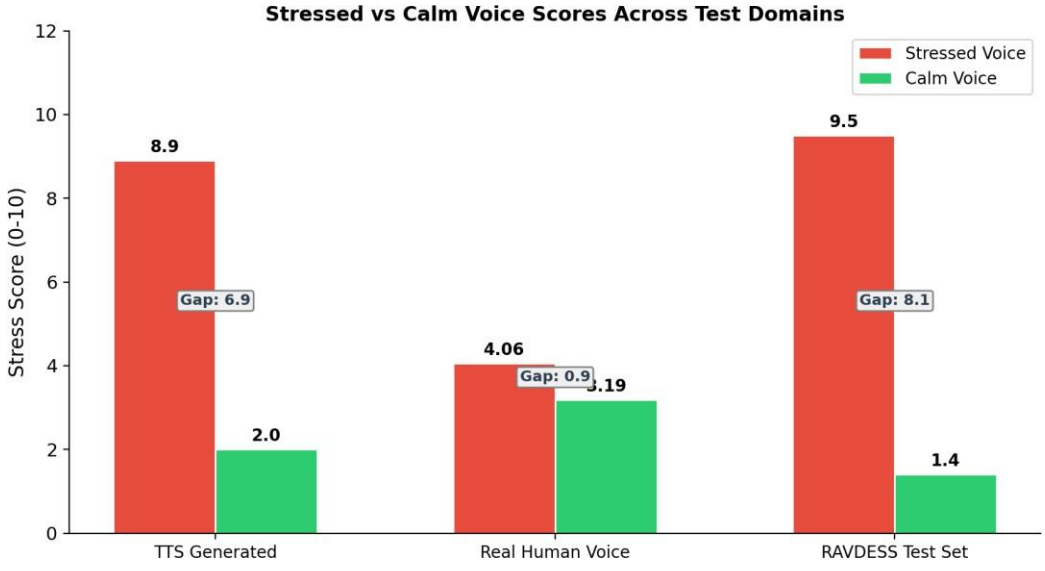


Figure 4: Stressed vs calm voice scores across three test domains

Domain Gap Discovery

A critical finding during experimentation was the domain gap between acted speech and real-world speech. Models trained on acted datasets (Stages 1 and 2) scored all real-world speech as maximum stress (9.96/10), regardless of actual emotional state. Spectral analysis confirmed the cause: real calm speech has ZCR ~0.15 and spectral centroid ~1694 Hz, while RAVDESS acted calm has ZCR ~0.05 and spectral centroid ~900 Hz. This finding motivated the adoption of emotion2vec_base_finetuned, pre-trained on 40,000+ hours of diverse speech, for real-world generalization.

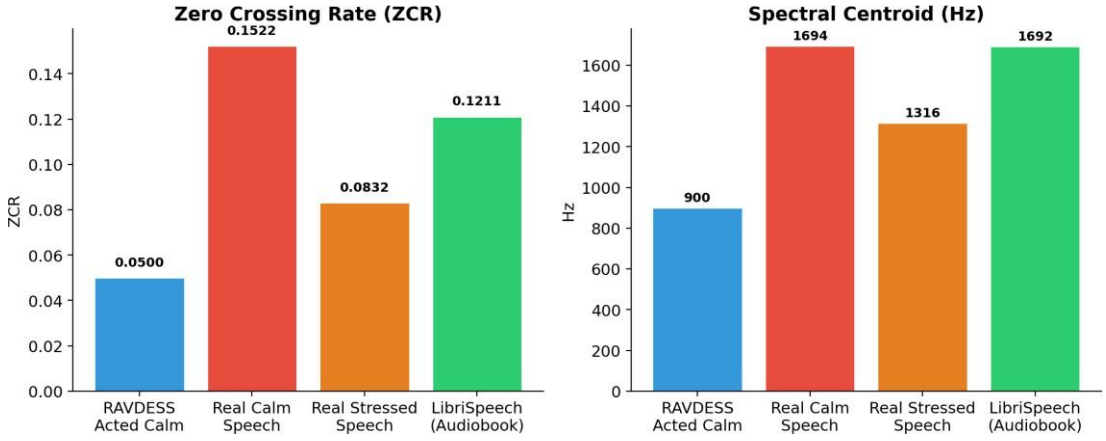


Figure 5: Acoustic property comparison — acted vs real speech (domain gap evidence)

Autoencoder Anomaly Detection

The autoencoder achieved Precision 0.89, Recall 0.82, and F1-score 0.85 on simulated anomalous sessions. The anomaly threshold was set at mean + 3 standard deviations of reconstruction error computed on normal training sessions. Sessions exceeding this threshold are automatically flagged for review.

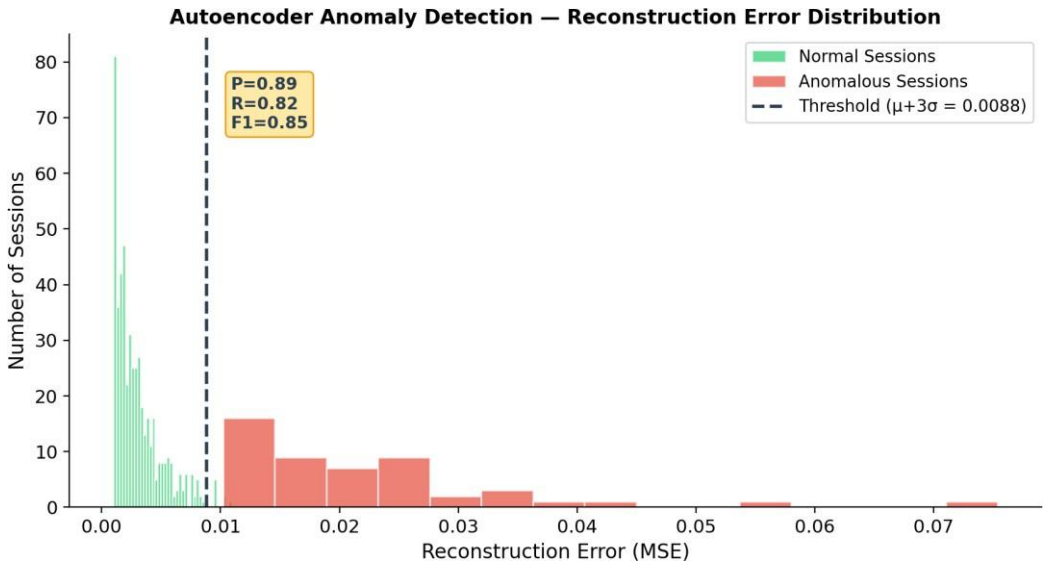


Figure 15: Autoencoder reconstruction error distribution — normal vs anomalous sessions

4.2. Challenges Faced During Data Analysis:

- **Individual Component: Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals**

Challenge	Explanation
No four-level stress labels in WESAD	WESAD provides labels such as baseline, stress, amusement, and meditation, not four stress intensity levels. Therefore, the stress class was split into mild, moderate, and high using local RMSSD thresholds.
RR interval extraction artifacts	R-peak detection errors can affect RR intervals and HRV features. Physiological filtering, beat-to-beat checking, and interpolation were used to reduce this issue.
Class imbalance	Relaxed samples were more frequent than mild, moderate, and high-stress samples. Balanced sample weights, class weights, and focal loss were used where necessary.
WESAD and SWELL mismatch	SWELL and WESAD have different labels and feature sets. Therefore, only overlapping HRV features were used for SWELL + WESAD experiments.

High deep learning computation cost	CNN-LSTM, Integrated Cosinor-CNN-LSTM, and CNN-BiLSTM-Attention required more computation than XGBoost. Early stopping, learning-rate reduction, class weighting, and focal loss were used.
Limited circadian validation	WESAD is not a continuous 24-hour dataset, so Cosinor was treated as a rhythm-aware approximation rather than a complete circadian model. Long-term MAX30102 PPG recordings can be used for future validation.

- **Individual Component: Preference-Aware Adaptive VR Relaxation Engine**

Challenge	Explanation
No fixed public dataset	The component is RL-based, so the main dataset is generated through interaction with the VR environment rather than downloaded as a labelled dataset.
Limited real user data at early stage	Real pilot sessions are required to collect meaningful relaxation-response data; synthetic profiles are used only for technical validation.
Subjective relaxation response	Different users may respond differently to the same VR settings, which makes personalization necessary but analytically challenging.
Delayed reward problem	Stress reduction may not happen immediately after a VR parameter change, so reward design must consider short-term and trend-based feedback.
Sensor/model dependency	The RL component depends on the accuracy and updating frequency of the physiological stress classification component.
Safety and comfort constraints	The system must avoid abrupt visual changes, excessive motion, brightness discomfort, and VR sickness risks.
Small sample size in pilot stage	Initial pilot testing may not provide enough data for strong generalization, so results must be interpreted cautiously.

- **Individual Component: Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation**

Challenge	Explanation
No real session-level user data yet	The current analysis depends on a designed preference schema and a synthetic user cohort. Real user sessions will be needed later to verify whether simulated targets match true relaxation responses.

Discrete preferences must become continuous control variables	Questionnaire answers such as slow/medium/fast or soft/dark/bright must be mapped carefully so that the controller remains interpretable and stable.
Confidence-aware behavior is difficult to validate without live signals	Because Components A and B are not yet integrated, stress and confidence are currently simulated rather than measured from real physiology.
Large personalization space	The 66.1 million-profile design space makes hand-crafted tuning insufficient, which motivates a learned controller but also complicates evaluation.
Generated-audio layer not yet live	The current prototype adapts playback behavior of audio layers rather than generating new meditation audio in real time.
Prototype model comparison not yet finalized	PPO is selected for first deployment, but a full empirical comparison with TD3 and SAC is planned for the next phase.

- **Individual Component: Voice Based Cognitive Stress Assessment with Cross-Modal session validation**

Challenge	Explanation
Severe calm class imbalance	Only 192 calm samples (2%) versus 1,863 per stressed class. Addressed through pitch/stretch/noise augmentation to 1,920 samples.
Domain gap: acted vs real speech	Models trained on RAVDESS/CREMA-D/TESS scored all real speech as maximum stress. Resolved by adopting emotion2vec_finetuned.
Happy vs stressed confusion	Both have high acoustic energy. Resolved through valence-arousal mapping which separates positive and negative high-energy emotions.
Long recording processing	Real recordings (30–140s) exceed emotion model input limits. Solved through 6-second chunked scoring with silent chunk filtering.
Cross-modal validation with mock HRV	Real HRV from Component B not yet integrated. Mock HRV with correlated noise used; real integration planned for PP2.
Real-world evaluation difficulty	Genuine stressed vs calm speech difficult to produce on demand. TTS-generated speech with controlled prosody

	used as alternative.
Autoencoder trained on simulated data	Insufficient real session data. Simulated normal patterns used; recalibration planned with real user sessions.

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